

RoboNeuro Mapping

Abstract

Inspired by mammalian hippocampal spatial cognition, this study develops a bio-inspired reinforcement learning architecture for autonomous robotic navigation. Traditional navigation methods such as SLAM face challenges in dynamic environments and computational complexity. By incorporating biologically-inspired mechanisms like boundary vector cells (BVCs), head direction cells, and place cells, our approach overcomes these limitations. Specifically, we integrate vision embedding via EfficientNetV2, a convolutional neural network, to fuse LiDAR-derived BVC representations with rich visual data.

We introduce a fusion layer combining visual embeddings (1280-dimensional vectors from EfficientNetV2) and spatial representations from the BVC output, normalized and processed into a coherent 400-dimensional tensor. Our experiments utilize biologically realistic learning rules—OJAS and HEBB—to evaluate spatial representation quality and navigation performance within simulated environments. Experimental results demonstrate significant improvements with vision embedding, producing more distinct, brighter, and spatially concentrated place fields, reducing aliasing, and enhancing navigational accuracy. This approach provides robust context-aware spatial cognition, effectively distinguishing visually similar yet semantically different locations, thereby advancing autonomous robotic navigation capabilities.

Introduction

Background

Inspired by the spatial cognition of mammals, particularly the hippocampal place cells, this project builds upon a biologically inspired model for robot navigation. Traditional SLAM systems struggle with dynamic environments and computational cost. In contrast, hippocampal models offer a lightweight yet adaptive alternative^{1,2}.

The original model mimics how animals use spatial cells—like boundary vector cells (BVCs), head direction cells (HD), and place cells (PCs) — to build cognitive maps. It uses Hebbian learning and competitive plasticity to form localized “place fields.” Navigation is driven by reward-based reinforcement through replay of prior episodes³. However, it originally relied solely on LiDAR input, limiting perception richness compared to biological systems.

Prior to our involvement, the original model demonstrated:

- Formation of convex place fields driven by BVCs.
- Goal-directed navigation using a replay-based reinforcement learning mechanism.
- Simulation-tested performance in Webots R2025 environments with rapid learning and stable path formation after minimal training⁴.

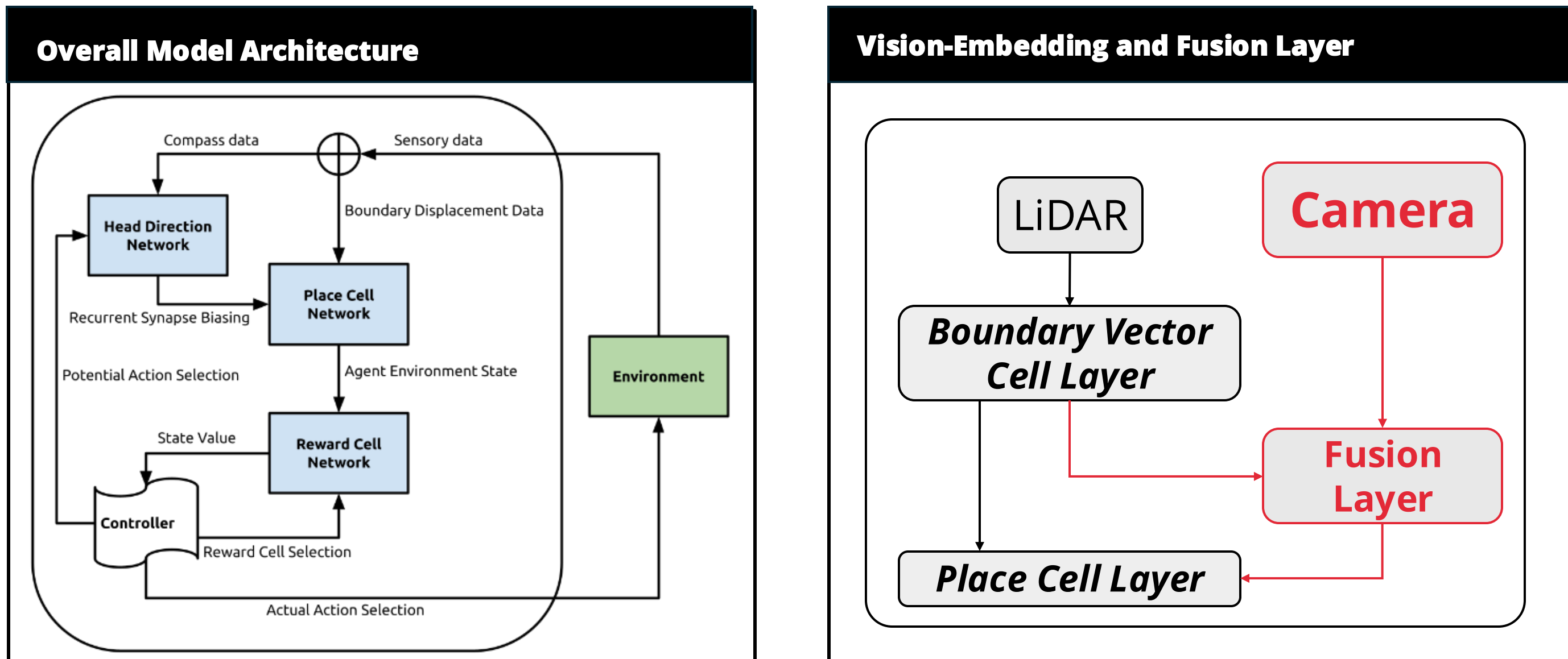
Mathematically, the place cell activation v_i^p is modeled as:

$$v_i^p = \tanh([\psi s_i^p]^+)$$

where s_i^p is a weighted sum of excitatory BVC input and lateral inhibition. Weight updates follow a form of Oja's rule:

$$\frac{dW_{ij}^{pb}}{dt} = v_i^p (v_j^p - \frac{1}{\alpha} v_i^p W_{ij}^{pb})$$

where v_j^p is the BVC firing rate, W_{ij}^{pb} is the synaptic weight from BVC j to place cell i , and α is a normalization factor. These principles resulted in robust spatial representations with minimal exploration time.



A bio-inspired reinforcement learning architecture integrating head direction, place cell, and reward cell networks for autonomous navigation.

The red section represents our vision embedding model, which fuses BVC outputs and camera inputs before feeding them to the place cell layer.

Objective

Our goal is to augment the model with visual perception, enabling richer, context-aware spatial learning. By fusing the camera and LiDAR data, we aim to:

- Enhance place field quality and reward propagation.
- Distinguish between visually similar but semantically different spaces⁵.
- Improve navigation performance under ambiguity or aliasing.

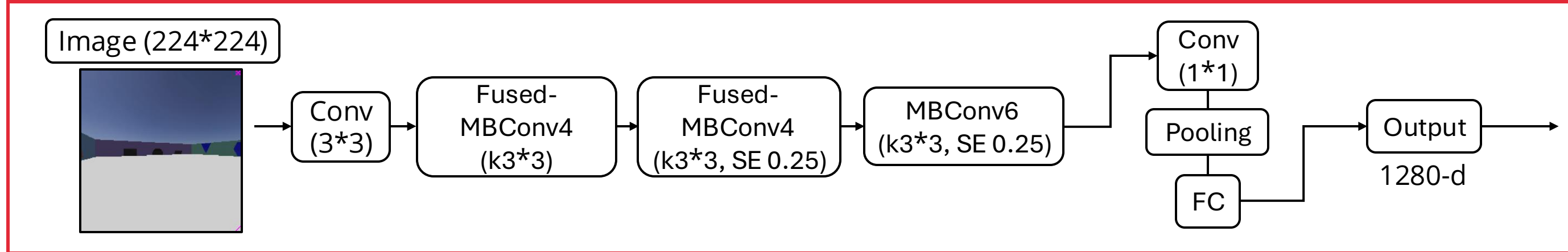
We integrate visual embeddings, which are CNNs named *EfficientNetV2* into the place cell layer and test multi-sensory fusion strategies like addition and concatenation⁶.

The upgraded model is deployed on the Webots R2025 platform and evaluated through spatial heatmaps, reward activation, and navigation efficiency.

Methodology

Vision-Embedding

We use EfficientNetV2B0 (initialized with ImageNet weights) as a feature extractor by removing the top classification layer and applying global average pooling, producing a 1280-dimensional vector. As shown in the diagram, the network comprises an initial 3×3 convolution, followed by Fused-MBConv and MBConv6 blocks with depthwise separable convolutions, and ends with a global average pooling layer. This final pooling outputs the 1280-length feature vector, which is then converted to a PyTorch tensor.



EfficientNetV2 Model Architecture Diagram.

Fusion Layer

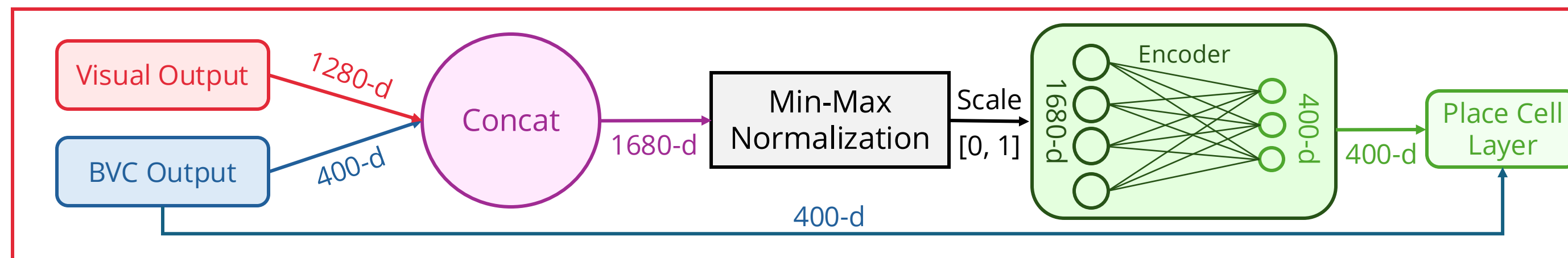
We introduced a Fusion layer to integrate the BVC output with the visual model output. The dimension of the BVC output is defined by the product of the number of head directions and the number of cells per direction—in this experiment, 8 heads multiplied by 50 cells per direction. The penultimate layer of the EfficientV2 model produces a 1280-dimensional output. These two outputs are first directly combined to form a 1680-dimensional tensor, which is then normalized using min-max normalization before being fed into a pretrained encoder for information fusion, resulting in a 400-dimensional fused information tensor that has the same dimension as the original BVC output.

$$\tilde{x} = \frac{(x_{BVC}x_{visual}) - \min((x_{BVC}x_{visual}))}{\max((x_{BVC}x_{visual})) - \min((x_{BVC}x_{visual}))}$$

Where $x_{BVC} \in R^{400}$ is the output from the BVC layer, and $x_{visual} \in R^{1280}$ is the visual model output.

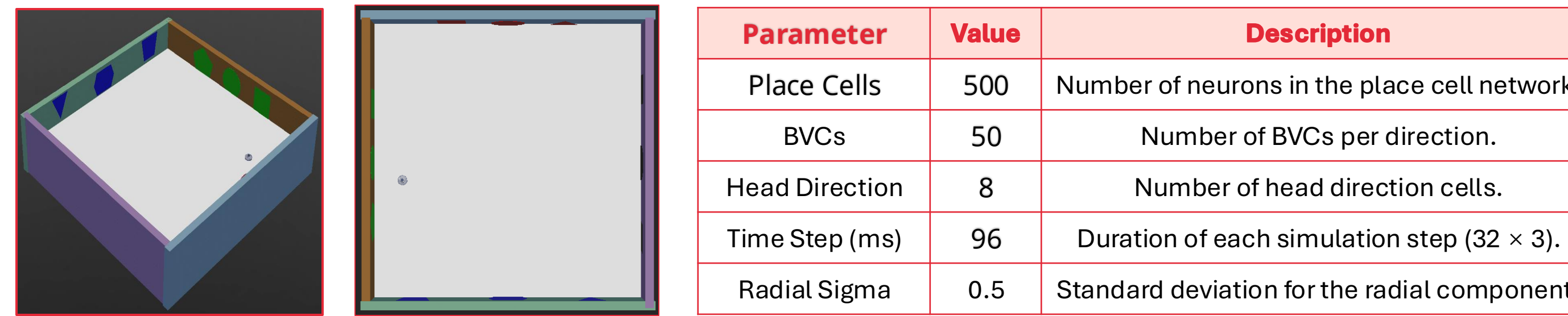
$$y = W \tilde{x} + b$$

Where $W \in R^{400 \times 1680}$ is the weight matrix of the encoder, and $b \in R^{400}$ is the bias vector.



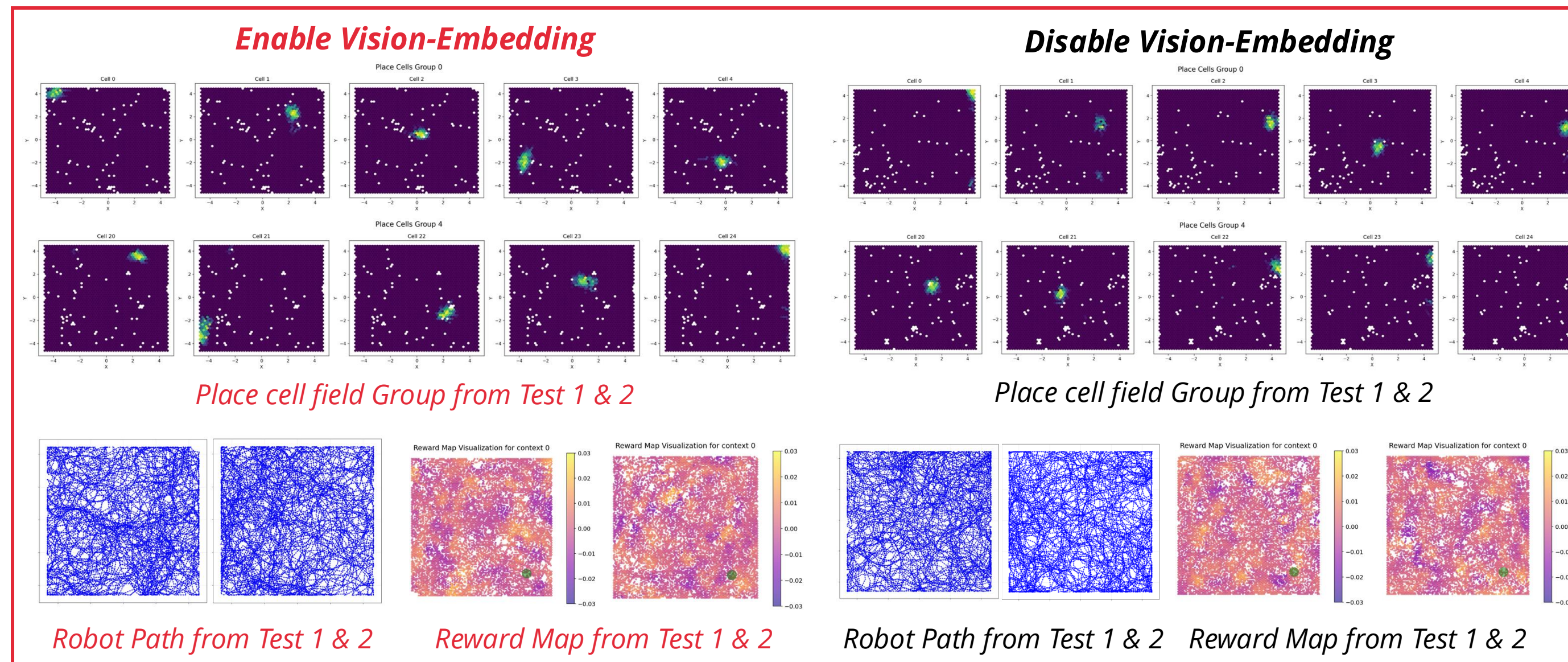
BVC and Visual outputs are passed through a fusion layer to generate a 400-dimensional fused feature vector.

Results and Discussion



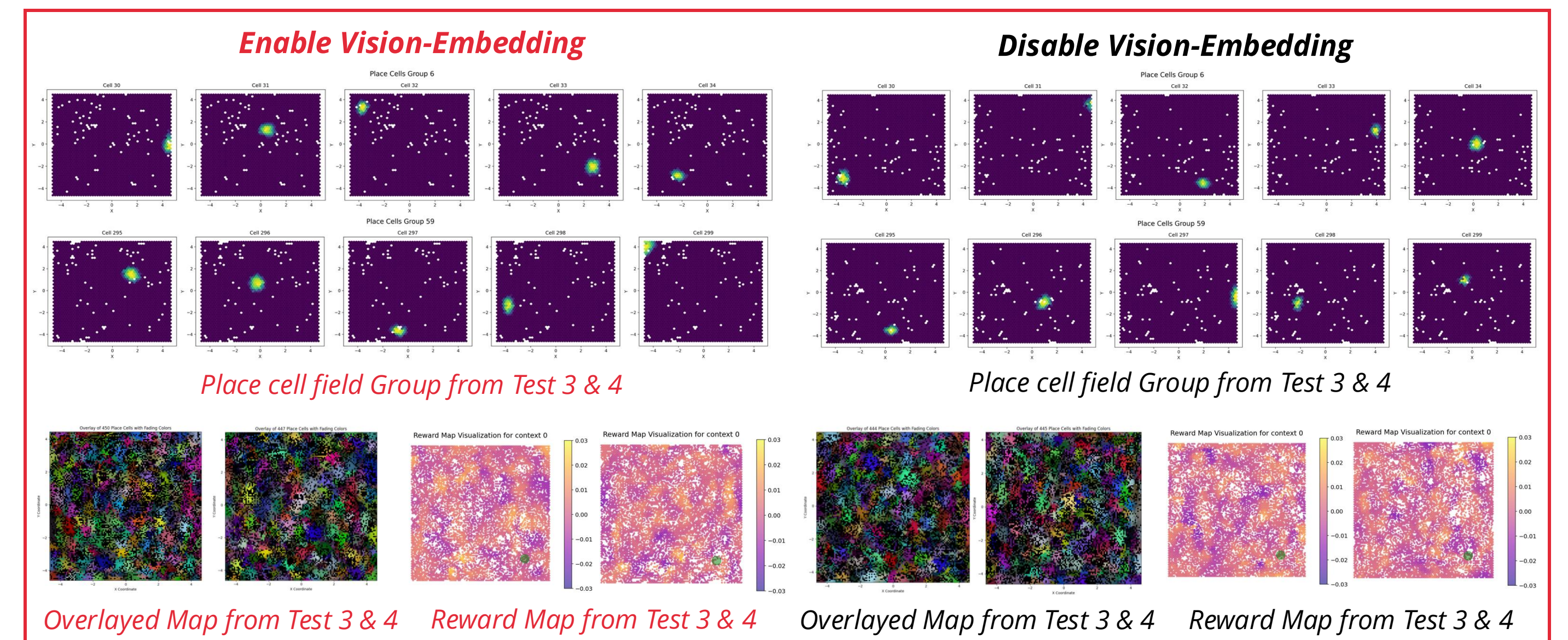
The test environment employs a 10×10 world, with each of the four walls featuring a unique color and pattern.

OJAS Mode



Comparison of Test Results: Enabling vs. Disabling Vision Embedding in OJAS Mode.

HEBB Mode



Comparison of Test Results: Enabling vs. Disabling Vision Embedding in HEBB Mode.

We evaluated two biologically inspired learning rules: OJAS and HEBB for training the place cell layer. OJAS mode emphasizes competition and decorrelation, producing broad, evenly distributed place fields. This supports faster spatial generalization, especially in novel environments. HEBB mode, by reinforcing co-activated synapses, forms sharper and more localized place fields. While this enhances spatial specificity and goal focus, it may also lead to redundancy or overfitting in some regions.

Importantly, our visual analysis goes beyond simply counting white (inactive) areas. The key metric is the quality of place field formation: With vision embedding, place cells show stronger, brighter activations (yellow-green regions) that are more spatially distinct and non-overlapping.

This indicates that individual place cells are encoding specific regions more clearly, leading to reduced aliasing and improved spatial resolution. Without vision, place fields are weaker, less defined, and more likely to overlap—suggesting less efficient spatial encoding.

Conclusion

In Conclusion, our vision-augmented place cells exhibit brighter, more concentrated, and spatially well-separated activation patterns, supporting stronger spatial representation and reward learning. Future work may explore hybrid strategies, combining the generalization ability of OJAS with the specificity of HEBB to further optimize navigation performance.

References

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